

Title: Analysis of the MediSecure Data Breach – A Real-World Information Security Incident

1. Overview of the Incident

In April 2024, MediSecure, an Australian electronic prescription service provider, experienced a significant ransomware attack. The breach resulted in the exfiltration of approximately 6.5 terabytes of data, compromising the personal and health information of about 12.9 million individuals. The stolen data included names, dates of birth, addresses, contact information, Medicare numbers, healthcare identifiers, prescription details, and concession card numbers. MediSecure confirms that approximately 12.9 million Australians who used the MediSecure prescription delivery service during the approximate period of March 2019 to November 2023 are impacted by this Incident based on individuals' healthcare identifiers. However, MediSecure is unable to identify the specific impacted individuals despite making all reasonable efforts to do so due to the complexity of the data set.

2. Analysis of the Security Attack

i. Type(s) of Security Attack

- **Ransomware Attack:** The primary attack involved ransomware that encrypts MediSecure's database servers, rendering them inaccessible. Ransomware is a form of malicious software designed to block access to a computer system or data, typically by encrypting files, until a ransom is paid. These attacks have become increasingly sophisticated and prevalent, posing significant threats to individuals, businesses, and critical infrastructure worldwide that also affect Medicare.
- **Data Exfiltration:** Prior to encryption, attackers exfiltrated a substantial amount of data (6.5 TB), indicating a sophisticated breach aimed at both disruption and data theft. Data exfiltration is a critical cybersecurity threat involving the unauthorized transfer of data from a computer or network to an external destination. This process can be executed manually by individuals with physical access or automatically through malicious software. The primary objective is to steal sensitive information such as personal data, financial records, or intellectual property.
- **Third-Party Vulnerability Exploitation:** The breach is believed to have originated from a third-party vendor, highlighting the risks associated with supply chain vulnerabilities.

ii. Impact on the Affected Organisation

- **Operational Disruption:** MediSecure was forced to shut down its website and phone lines to contain the attack, leading to significant service disruptions.
- **Data Loss and Inability to Identify Affected Individuals:** Due to the complexity and volume of the data, MediSecure was unable to identify specific individuals affected by the breach. The data was stored in semi-structured and unstructured formats across various datasets, making it impractical to pinpoint affected individuals without incurring substantial costs that the company could not afford.
- **Financial and Reputational Damage:** The breach led to significant financial strain, culminating in MediSecure entering voluntary administration. The incident also damaged the company's reputation and trust among stakeholders.

3. Proposed Countermeasures

To prevent or mitigate similar incidents, the following countermeasures would be considered:

a. Implement Robust Third-Party Risk Management

Establish stringent security requirements and regular audits for third-party vendors to ensure they adhere to security best practices, reducing the risk of supply chain attacks. Keep all software up to date, especially vulnerable applications like Flash, to eliminate exploitable security holes. Next, it's crucial to install and regularly update reputable antivirus software to detect and block malicious threats before they can execute. An essential safeguard is to perform regular backups of important data to secure external locations such as USB drives, external hard disks, or cloud storage. This allows data recovery without giving in to ransom demands if an infection occurs.

b. Enhance Data Encryption and Access Controls

Encrypt sensitive data both at rest and in transit. Enhanced database security relies on strong access controls (DAC, MAC, RBAC, ABAC) and encryption (symmetric, asymmetric). These techniques restrict unauthorized access and protect sensitive data by converting it into unreadable formats. Encryption can be applied at storage, database, or application levels, ensuring confidentiality even if breaches occur.

c. Develop and Test Incident Response Plans

Create comprehensive incident response plans that include regular drills and updates. This ensures preparedness to respond swiftly to breaches, minimizing damage and recovery time. The

management and security should build the response plan and implement it while facing any incident that could potentially harm the organization so that they can prevent it before occurring.

4.Article : 12.9 million individuals affected by MediSecure cyber breach

<https://www.securitymagazine.com/articles/100875-129-million-individuals-affected-by-medisecure-cyber-breach>

5.References:

1. Nelson, S. D., & Simek, J. W. (2020). Ransomware as a data breach: An evolving threat. *Law Prac.*, 46, 24.
https://heinonline.org/hol-cgi-bin/get_pdf.cgi?handle=hein.journals/lwpra46§ion=58
- 2.Liu, Y., Corbett, C., Chiang, K., Archibald, R., Mukherjee, B., & Ghosal, D. (2009, January). SIDD: A framework for detecting sensitive data exfiltration by an insider attack. In *2009 42nd Hawaii international conference on system sciences* (pp. 1-10). IEEE.
<https://ieeexplore.ieee.org/abstract/document/4755414/>
- 3.Hong, S., & Yu, J. (2018). Ransomware attack analysis and countermeasures of defensive aspects. *Journal of Convergence for Information Technology*, 8(1), 139-145.
<https://koreascience.kr/article/JAKO201809538046711.page>
4. Khalaf, E. F., & Kadi, M. M. (2017). A survey of access control and data encryption for database security. *Journal of King Abdulaziz University*, 28(1), 19-30.
http://www.kau.edu.sa/Files/320/Researches/70647_43621.pdf
5. Farok, N. A. Z., & Zolkipli, M. F. (2024). Incident response planning and procedures. *Borneo International Journal eISSN 2636-9826*, 7(2), 69-76.
<http://majmuah.com/journal/index.php/bij/article/view/641>